INTERNSHIP REPORT ON BUSINESS INTELLIGENCE TOOL NAMED TABLEAU

INTERN'S NAME

MD ABDUR RAHMAN
International FE 2022
Card Number: TTU 9260

SUPERVISOR

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ORGANIZATION

International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b)

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Acknowledgement

In the name of Allah, the Most Gracious, the most merciful. First and foremost, praises to Allah the Almighty for His countless of blessings throughout my internship journey to complete my Data analyst internship Programme 2022 with ease. The Almighty Allah who granted me health and long life, without which I could not have finished this internship successfully.

I would like to express my deep and sincere gratitude to my supervisor, Dr. Md. Sharful Islam Khan, Scientist and Head, HIV & AIDS Programme, Infectious Diseases Division for giving me the chance to undergo my Data analyst internship in this Programme under IDD. He had the kindness to accept me in his Programme and guide me through my internship with his advice and feedbacks despite his busy schedule. Dr. Md. Sharful Islam Khan is compassionate, and he has consistently assisted me in grabbing the chance to go further into my experience as a data analyst intern in Bangladesh.

Not to mention, my thankfulness to Mr. Masud Reza as Sr. Manager M&E, HIV & AIDS Programme, IDD at the icddr,b for assisting me and directing me throughout this internship time. Also, my gratitude to Mr. Dr A.K.M. Masud Rana Project Coordinator, HIV & AIDS Programme, Infectious Diseases Division at the icddr,b for observing my tasks frequently during my internship.

My thankfulness to Mr. Mohammad Sha Al Imran, Assistant Programme Manager Data M&E, HIV & AIDS Programme, IDD at the icddr,b for providing information about Microsoft excel, recommending about graphs and so on. Moreover, among other things, my profound appreciation to Mr. Khandaker Waliur Rahman as Data Management Supervisor, HIV & AIDS Programme, IDD at the icddr,b for supporting and guiding me during this internship period. Preparing data files and documents, continuous questions, and inquiries, and helping to connect with the data team, all these tasks are done with the help of his without any complaint or objections.

All my fellow employees and seniors deserve a special thank you for sharing their knowledge, effort, time, and enthusiasm, especially throughout this internship program. Having the opportunity to meet so many lovely individuals and experts who guided me during this internship time makes me thankful as well. I appreciate your continued encouragement as I work toward my goals.

Last but not least, I would like to thank my biggest support in life, my parents. The covid pandemic has affected me a lot, mentally and physically. It was kind of hard for me to be motivated and enthusiasm in everything that I am doing. Without them, I do not know what kind of person I would be at this moment. Thanks to them, they are the reason for who I am and where I am standing today. May Allah grant them happiness, barakah and always be showered with prosperity. I also pray that Allah the Most Merciful will give them long life and health so that I can pay them back as much as I could as a child and serve them till the last breath of my life. In sha Allah.

Intern's details

Student's Name: **Md Abdur Rahman**
Matric Number: **1639233**
Program: **Business Intelligence and information science**
University: **International Islamic University Malaysia**
University Address: **Jalan Gombak, 53100, Selangor, Kuala lumpur, Malaysia**

Education

- Currently studying as a BIT student (4th year)
  International Islamic University Malaysia (IIUM) (2019 - Present)
- High School (Science) with a GPA of 5.00 out of 5
  Tamirul Millat Kamil Madrasah, Tongi, Bangladesh
- School (Science) with a GPA of 5.00 out of 5
  Tamirul Millat Kamil Madrasah, Tongi, Bangladesh
- Junior School with a GPA of 4.89 out of 5
  Tamirul Millat Kamil Madrasah, Tongi, Bangladesh

Internships

- Currently enrolled as a junior data analyst for 2 months hybrid internship with icddr,b *(Card Number: TTU 9260)*
- Successfully completed 2 months of Virtual Internship with KPMG Australia. (Enrolment Verification Code: sd7ZEfTCXECtYWute, User Verification Code: 5Saec8LF4XQQDHMHT | Issued by Forage)
- Successfully completed the 5-week Internship with International MUN. (Batch: Forum - Al, Referral code: KK1153)

Projects

- Currently, working on a project using JavaScript, and React Native. (Final year Project)
- I have developed several projects using HTML, CSS, JavaScript, Laravel, and PHP. (Client project)
- Zakat calculation website. (Academic project)
- Personal website for a real-time client.
- 2nd-hand educational material's website. (Academic project)
- Health-Care-related website by using HTML, CSS, and JavaScript. (Academic project)

Acronym

DGHS - Directorate General of Health Services

FSW - Female sex worker

GF - Global Fund

HIV - Human immunodeficiency virus

KP - Key populations

KPI – Key performance indicator

MIS - Management information system

MSM - Men who have sex with men

MSW - Male sex worker

NGO - Non-governmental organization

OST – Oral Substitute Therapy

PLHIV - People living with HIV

PMTCT - Prevention of mother-to-child transmission

PWID - People who inject drugs

STI - Sexually transmitted infection

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Introduction

The reason why I chose this organization is that I can see icddr,b is a uniquely placed to generate a better understanding of the most pressing health problems facing Bangladesh and the global South, and to develop effective solutions to those problems. icddr,b running multiple division, Program for HIV and AIDS under the direction of the Infectious Diseases Division (IDD) is one of them and which are working with Men who have sex with men (MSM), Male sex worker (MSW), HIJRA (transgender women) & People who inject drugs (PWID). Moreover, since 2010, the Global Fund has collaborated with the Program for HIV and AIDS, under the direction of the Infectious Diseases Division (IDD) of icddr,b. The project funded by the Global Fund aims to reduce the risk of HIV transmission in Bangladesh's major populations. Since HIV & AIDS division is working with those population, HIV & AIDS division has been collecting huge amounts of data to analyse the target population of HIV & AIDS. So, I had been assigned to work with Gender based violence (GBV) & Human rights (HR) data and it is required to develop a dashboard for GBV & HR data visualisation.

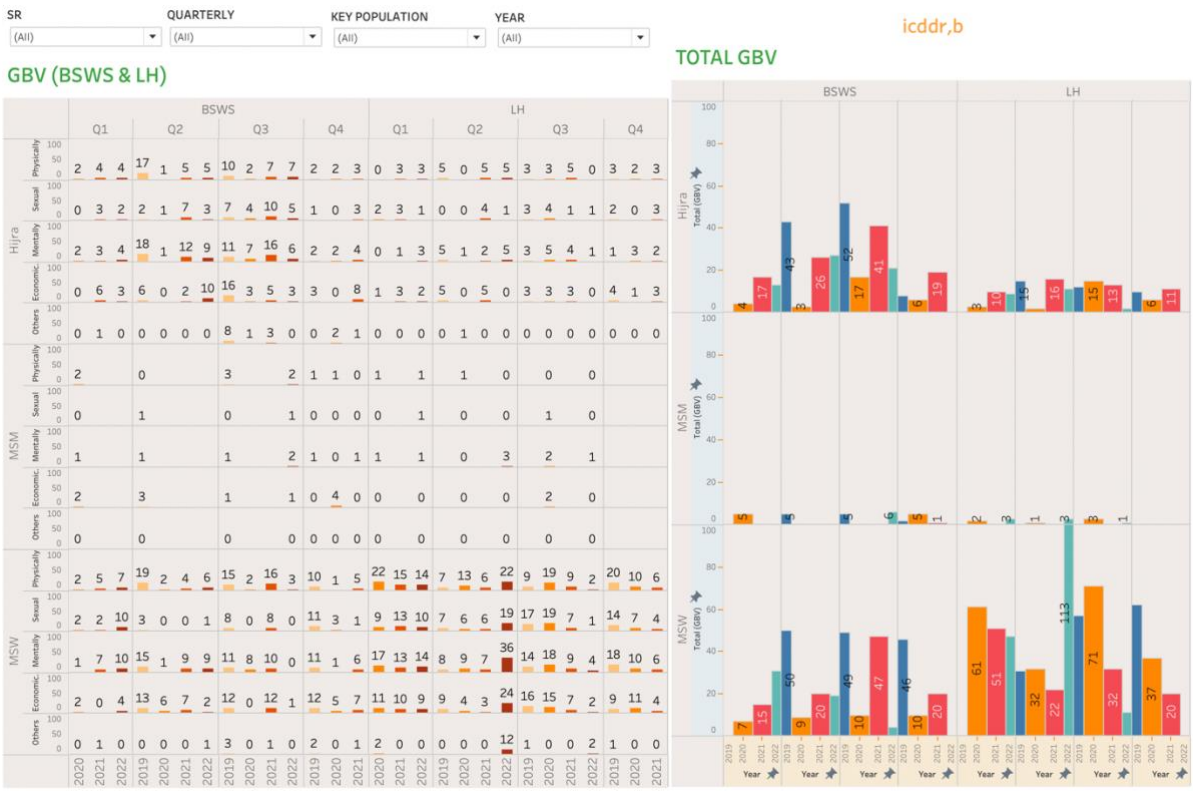


Figure 1: GBV dashboard from April 2019 to July 2022

Above picture illustrates GBV dashboard from 2019 to 2022. User can view all GBV data by SR, yearly, quarterly, and key population.

Total of 632 cases of GBV and HR were discovered between April 2019 and July 2022. Five incidences of gender-based violence—physical, sexual, economic, mental, and others—are highlighted in the excel file, while eight incidences of HR—no HR, privacy, human & personal security, adequate standard of living, education, freedom of expression, opinion, assembly, and

association, highest standard of health, and participation in cultural and family life—are also highlighted. There are four filters available from the dashboard to fully comprehend the GVB data on Tableau. Service recipient (SR), which includes Bondhu and the lighthouse, is the first filter. Moreover, users can access the data every three months using a filter named Quarterly. Additionally, the dashboard users may observe each of the three main populations that have been highlighted. Finally, users may discover every year in year filter and utilize dashboard to see and comprehend data more effectively.

Organization information

icddr,b was founded in 1960 with the intention of Research on Cholera. The research centre's mission is to solve public health problems through innovative scientific research. icddr,b plays a vital role in the community in solving public health problems facing low- and middle-income countries through innovative scientific research – including laboratory-based, clinical, epidemiological, and health systems research. icddr,b is one of the world's leading global health research institutes. Based in Dhaka, Bangladesh. That's why I chose this organization because I find its mission to be essential and relevant.

Brief about HIV & AIDS Programme

The HIV and AIDS program at icddr,b has collaborated with numerous community-based organizations to gain direct exposure to populations at risk for contracting HIV and to offer them high-quality sexual health and other services. It has also worked with the government of Bangladesh to produce strategic data that will help programs. This dedication to fostering relationships reflects the extensive national and international networking that has significantly lowered the HIV pandemic in Bangladesh. Currently, 50 health facilities constructed throughout 37 districts in cooperation with the Bandhu Social Welfare Society (BSWS) and Light House (LH) provide HIV prevention services to almost 37,500 sexual minority persons and 5 OST centre's operated by icddr,b for 600 clients.

Justification

HIV & AIDS receive GBV & HR data from service recipient (SR) in excel format afterward they need to process the data for better understanding. HIV & AIDS presents data by using excel graphs like bar charts, line graphs, pie charts, and so on. Processing and presenting data can occasionally be time-consuming. In order to allow the service recipient (SR) to input data from Google Sheets, Excel, and MySQL, I have developed a dashboard in Tableau. Additionally, tableau allows customers to visualize the data or the dashboard in as many different ways as they desire.

Challenge & limitation

At icddr,b, this is the first time I've had the opportunity to work with actual office employees. Although this is the ideal opportunity for me, there were some difficulties. For instance, as an

intern, I wasn't used to talking to other workers about the data or using certain terminology that aren't commonly used outside.
On the other hand, I was not able to access the data in my laptop since it was in Bangla, a language that my laptop does not support. Will I be able to work here, I worried. After a short while, I received a laptop, but in order to use it for work, it has to be fixed. When I finished the windows, everything ran nicely.

Consideration & recommendation

Well, I had a great experience with the company, but there were a lot of issues I had to deal with. For instance, problems with laptops, data processing, and other things are currently present but will change in the future. I would suggest adopting Tableau in the HIV & AIDS Program as it is one of the top business intelligence tools today and is used by the majority of the world's largest companies.

Internship description & project assigned

My role at icddr,b was a junior data analyst and I worked directly with M&E, MIS & Data team. By using Business intelligence tool named tableau, I had developed an interactive dashboard to represent visually GBV & HR data from April 2019 to July 2022.

Lesson learns

I. Learn basic info about HIV & AIDS

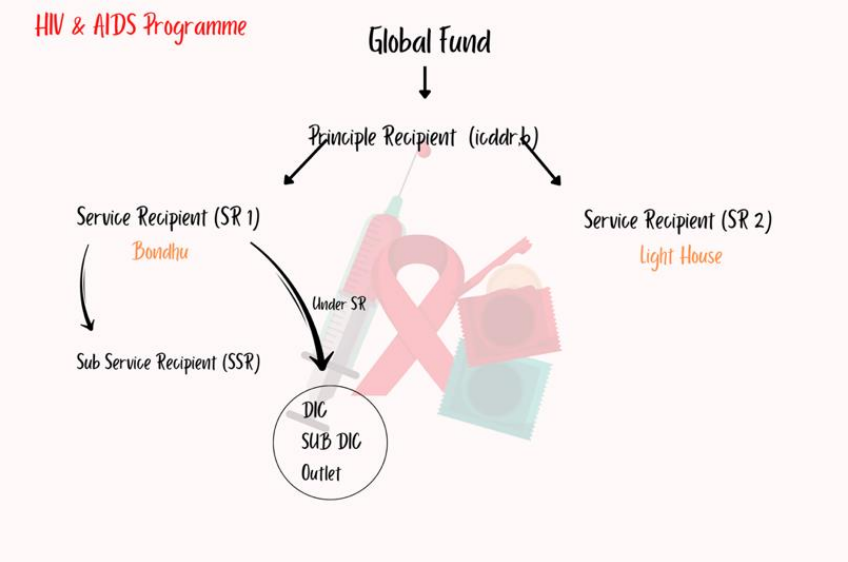


Figure 2: The picture illustrates how Global Fund work with icddr,b from initial to the end.

II. Acronyms: these Acronyms have been learned which is related with HIV & AIDS

Below are some significant acronyms that are connected to HIV and AIDS that everyone should be aware of.

DGHS - Directorate General of Health Services
FSW - Female sex worker
GF - Global Fund
HIV - Human immunodeficiency virus
KP - Key populations
KPI – Key performance indicator
MIS - Management information system
MSM - Men who have sex with men
MSW - Male sex worker
NGO - Non-governmental organization
OST – Oral Substitute Therapy
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PWID - People who inject drugs
STI - Sexually transmitted infection

III. Suggested tools to work with data

- KoboToolbox: (The most popular tool in humanitarian catastrophes is KoboToolbox, a free toolkit for gathering and organizing data in difficult environments)
- SPSS: (SPSS stand for statistical package for social science and it is utilised for the in-depth research in complex statistical data analysis)

IV. Some questions to know more about HIV & AIDS

- Q1 - What is HIV & AIDS? - HIV (human immunodeficiency virus) is a virus that attacks the body's immune system. If HIV is not treated, it can lead to AIDS (acquired immunodeficiency syndrome). There is currently no effective cure. Once people get HIV, they have it for life.
- Q2 - Who are the significant affected people in Bangladesh? – People how inject drugs (PWID), Male having sex with males (MSM), Male sex workers (MSW), Hijra (transgender women). Female sex workers.
- Q3 - How many people affected till now by HIV & AIDS in Bangladesh? – The estimated number of infections in Bangladesh is about 8,761 [1]
- Q4 - Any diseases with HIV & AIDS and what others name of them? – Pneumocystis pneumonia (PCP), Candidiasis (thrush), Tuberculosis (TB), Cytomegalovirus, Cryptococcal meningitis, Toxoplasmosis.
- World Aids Day – 1st December

V. Logistics Management Information Systems (LMIS)

- Learn about LMIS.
- How LMIS work in an organization.

- Opening Balance - The opening balance is the number of funds in a company's account at the beginning of a new financial period. It is the first entry in the accounts, either when a company is first starting up its accounts or after a year-end.
- Closing Balance - The debit or credit balance of a ledger account in the Chart of Accounts at the end of an accounting period or year-end is called the closing balance. This closing balance becomes the opening balance for the next accounting period.

VI. How icddr,b collects data from the targeted population?

DIC managers or peer educations collect data from the targeted population then they send the data to service recipients to verify the data. After verifying the data, SR (Service recipients) send the data to the principal recipients which is icddr,b.



Figure 3: The flow to collect data from targeted population.

VII. Some extra basic information about data

- Data => information => Knowledge
- Data can be number, label and so on.
- Information always will be organised (of data).
- Knowledge, use of information.

VIII. Info about MSM, MSW and HIJRA

- MSM – Males who have had sex with males or not they have sex with women or have a personal or social gay or bisexual identity but do not sell sex. (Total – 101,695) [2]
- MSW – Male who sell sex to other males in exchange of money or gifts. (Total – 29,777) [2]
- HIJRA – Those who identify themselves as belonging to a traditional hijra sub-culture and who maintain the guru-chela Hijra hierarchy. They maybe sub-

categorized as: Sex Worker Hijra, Badhaihijra and Radhunihijra. (Total – 10,199) [2]

IX. Book review

1. National Strategy Plan (NSP)
2. National HIV serological surveillance, 2004 – 2005
3. Mapping Study and Size Estimation of Key Populations in Bangladesh for HIV Programs 2015-2016
4. Behavioural and Serological Surveillance on males having sex with males, male sex workers and hijra, 2015.

X. Article review

1. Men who have sex with men's sexual relations with women in Bangladesh. Culture, health & sexuality 2005 [3]
2. Safer sex or pleasurable sex? Rethinking condom use in the AIDS era. Sexual Health 2005 [4]
3. Point of care HIV testing with oral fluid among returnee migrants in a rural area of Bangladesh. Current Opinion in HIV and AIDS 2016 [5]
4. HIV risk and preventive interventions in transgender women sex workers. The Lancet 2015 [6]
5. Sexually transmitted infections and sexual behaviour among youth clients of hotel-based female sex workers in Dhaka, Bangladesh [7]

XI. Relational Database

A relational database is a collection of data items with pre-defined relationships between them. These items are organized as a set of tables with columns and rows. Tables are used to hold information about the objects to be represented in the database.

XII. No SQL

NoSQL database technology stores information in JSON documents instead of columns and rows used by relational databases. To be clear, NoSQL stands for “not only SQL” rather than “no SQL” at all.

XIII. What is tableau and what is the main purpose of Tableau?

Tableau is an excellent data visualization and business intelligence tool used for reporting and analysing vast volumes of data. It is an American company that started in 2003 and in June 2019, Salesforce acquired Tableau.

Tableau enables individuals and businesses to become more data driven. This analytics platform, which is the industry leader for digital business intelligence, makes it simpler for users to explore and process information as well as faster to find and share insights that have the potential to transform organizations and entire civilizations.

XIV. Benefit of tableau

- Create Interactive Visualizations Quickly
- Convenient Implementation
- Easily Handles Large Amounts of Data
- The ability to use other scripting languages, and mobile friendly
- A large number of customer resources
- Exceptional Mobile Support & Affordable

XV. Learn about tableau

Tableau is a visual analytics tool that is revolutionizing how we utilize data to address issues and enabling individuals and companies to maximize their data.

XVI. Key Difference between Data Analysis and Data Analytics

Data analysis is a process involving the collection, manipulation, and examination of data for getting a deep insight. Data analytics is taking the analysed data and working on it in a meaningful and useful way to make well-versed business decisions.

XVII. Tasks

It is my obligation to deal with the GBV and HRV data, and I must construct or create an interactive dashboard on Tableau to display the GBV and HRV data.

XVIII. Process of work (Workflow)

This is the process flow of my work below,

- **Ask a question:** in this section, I has conducted a meeting to know about the data, like what type of data it is, what I need to do for this project.
- **Get the data**: GBV & HRV data has been collected from the Data, M&E team for this project.
- **Understanding data:** in this phase, I receive data and I start to understand the data for the project.
- **Explore data:** I start to explore the data in this phase. For example, data contents, location, time and so on.
- **Reformat & clean data:** in this phase, data has been fixing or removing incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data within a dataset.
- **Data preparation:** after cleaning & reformatting the data, data has been prepared for the project in this phase.
- **Developing dashboards by tableau:** required dashboards are developed by using GBV & HRV data in this phase.
- **Evaluation tasks:** dashboards are evaluated in this phase and check the interactive dashboards are work correctly.
- **Ready for use:** dashboards are ready to use in this phase.



Figure 4: work process (Workflow)

Overview of internship experience

During my internship experience with icddr,b, I was able to develop my data analytics skills by using business intelligence tool named tableau. I particularly found business intelligence tools experience to be useful in improving my data analysis skills. Although I found business intelligence tools to be challenging, I found it to be valuable in developing my data analysis skills.

Ongoing Consideration

While I had many useful experiences at icddr,b, I feel that I still need to develop my confidence levels in using business intelligence tools and data reprocessing method. I would have enjoyed more time completing data analysis skills.

Conclusion

I picked this organization because I believe it is well positioned to advance knowledge of the most important health issues confronting Bangladesh and the global South and to provide practical solutions to those issues. The Infectious Diseases Division (IDD) of the icddr,b is in charge of several divisions, among which is the Programme for HIV and AIDS. This division works with individuals who inject drugs, men who have sex with males, male sex workers, and transgender women (PWID). In addition, I was able to improve my data analytics abilities utilizing the business intelligence application tableau throughout my internship with icddr,b. Proficiency with business intelligence technologies was especially helpful to me in developing my data analysis abilities. Finally, I have spent my time with lots of experienced employee from here and I have gathered enough knowledge for my future career.

Appendix

- hivprogram.icddrb.org this web page has been used to work into this division.
- **How to clean unnecessary data from the laptop and how to make computer fast:** A few ways have been recommended to clear the laptop of extra files and speed up the machine. Most 2 major steps have been highlighted here. Step 1 - Go this PC => Select C Drive => Select Properties => Click Disk Clean and Step 2 - Always check for update.
- **Practical task - Install windows from 7 to windows 10**
    - A windows laptop has been given to do practical work. Device name – TOSIBA.
    - And requirements & suggestion – C Drive must contain 200 GB storage, must install windows 10 pro and some basic initial features need to be added into the device.
    - Security question and set up need to follow as instruction.

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